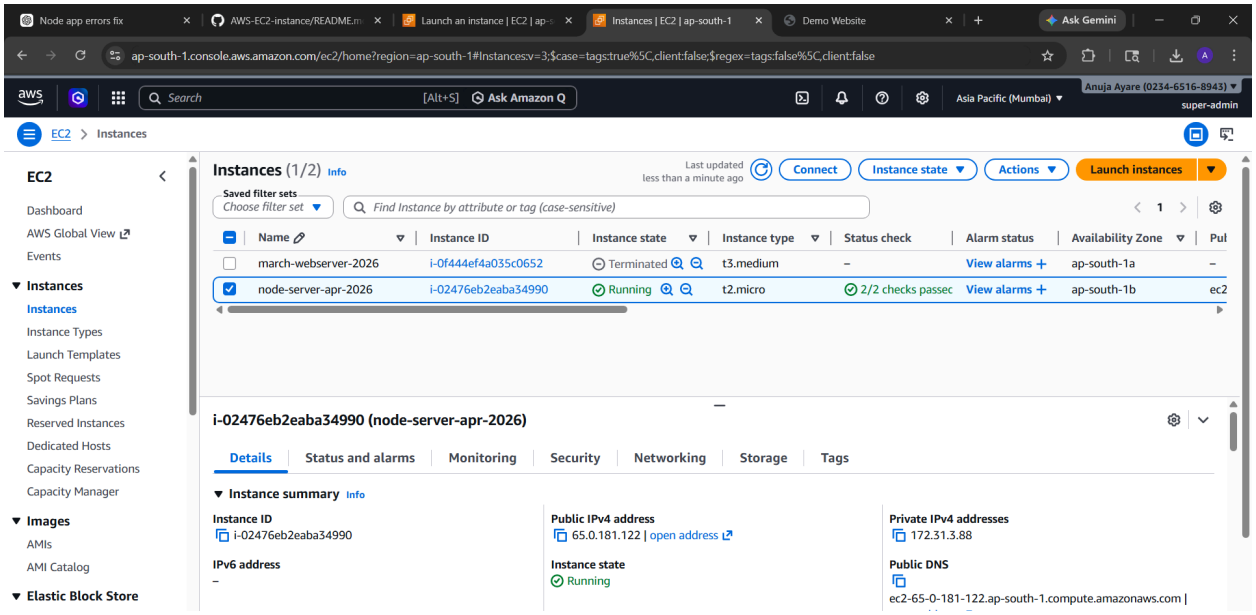
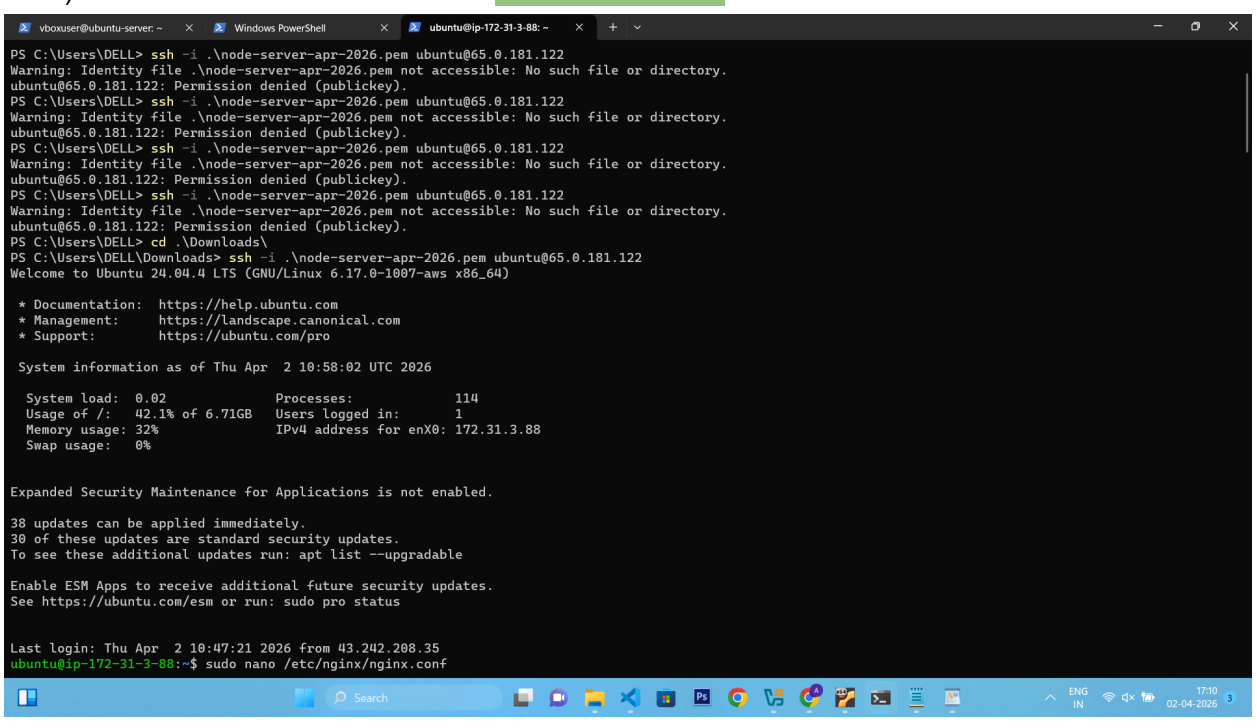
Deploying an Application on AWS EC2

1) Launch an EC2 Instance - Running EC2 Instance



2) Connect to the EC2 Instance : Server connected



3) Update and Install Dependencies : Installation Done

```
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
ubuntu@ip-172-31-3-88:~$ git clone https://github.com/ayareanuja0903-collab/Node_application.git
Cloning into 'Node_application'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 9 (delta 0), reused 6 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), done.
ubuntu@ip-172-31-3-88:~$ ls
Node_application
ubuntu@ip-172-31-3-88:~$ cd Node_application/
ubuntu@ip-172-31-3-88:~/Node_application$ npm install
Command 'npm' not found, but can be installed with:
sudo apt install npm
ubuntu@ip-172-31-3-88:~/Node_application$ sudo apt install npm
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu build-essential bzip2
  cpp cpp-13 cpp-13-x86-64-linux-gnu cpp-x86-64-linux-gnu dpkg-dev eslint
  fakeroot fontconfig-config fonts-dejavu-core fonts-dejavu-mono g++
  g++-13 g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu gcc gcc-13
  gcc-13-base gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu gyp handlebars
  javascript-common libalgorithm-diff-perl libalgorithm-diff-xs-perl
  libalgorithm-merge-perl libaom3 libasan8 libatomic1 libauthen-sasl-perl
  libbinutils libc-dev-bin libc-devtools libc6-dev libcc1-0 libclone-perl
  libcrypt-dev libctf-nobfd0 libctf0 libdata-dump-perl libde265-0
  libdeflate0 libdpkg-perl libdrm-amdgpu1 libdrm-intel1 libegl-mesa0
  libegl1 libencode-locale-perl libfakeroot libfile-basedir-perl
  libfile-desktopentry-perl libfile-fcntllock-perl libfile-listing-perl
  libfile-mimeinfo-perl libfont-afm-perl libfontconfig1 libgbm1
  libgcc-13-dev libgd3 libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-bin
  libgdk-pixbuf2.0-common libgl1 libgl1-mesa-dri libgles2 libglvnd0
  libglx-mesa0 libglx0 libgomp1 libgprofng0 libheif-plugin-aomdec
  libheif-plugin-aomenc libheif-plugin-libde265 libheif1 libhtml-form-perl
  libhtml-format-perl libhtml-parser-perl libhtml-tagset-perl
```

4) Deploy Your Application - Node_application

```
added 68 packages, and audited 69 packages in 3s

15 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
ubuntu@ip-172-31-3-88:~/Node_application$ node server.js &
[1] 7658
ubuntu@ip-172-31-3-88:~/Node_application$ Server running on http://localhost:3000
client_loop: send disconnect: Connection reset
PS C:\Users\DELL\Downloads> ssh -i .\node-server-apr-2026.pem ubuntu@65.0.181.122
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Apr  2 11:41:49 UTC 2026

  System load:  0.0               Processes:             115
  Usage of /:   43.2% of 6.71GB   Users logged in:       1
  Memory usage: 34%               IPv4 address for enX0: 172.31.3.88
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

38 updates can be applied immediately.
30 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Thu Apr  2 10:58:03 2026 from 43.242.208.35
ubuntu@ip-172-31-3-88:~$ cd Node_application/
ubuntu@ip-172-31-3-88:~/Node_application$ ls
README.md  node_modules  package-lock.json  package.json  public  server.js
ubuntu@ip-172-31-3-88:~/Node_application$
```

Server Running : http://localhost:3000

```
    at Module.require (node:internal/modules/cjs/loader:1225:19)
    at require (node:internal/modules/helpers:177:18)
    at Object.<anonymous> (/home/ubuntu/Node_application/server.js:1:17)
    at Module._compile (node:internal/modules/cjs/loader:1356:14)
    at Module._extensions..js (node:internal/modules/cjs/loader:1414:10)
    at Module.load (node:internal/modules/cjs/loader:1197:32)
    at Module._load (node:internal/modules/cjs/loader:1013:12)
    at Function.executeUserEntryPoint [as runMain] (node:internal/modules/run_main:128:12) {
  code: 'MODULE_NOT_FOUND',
  requireStack: [ '/home/ubuntu/Node_application/server.js' ]
}

Node.js v18.19.1
^C
[1]+  Exit 1                  node server.js
ubuntu@ip-172-31-3-88:~/Node_application$ sudo firewall-cmd --add-service=http --permanent
sudo firewall-cmd --reload
sudo: firewall-cmd: command not found
sudo: firewall-cmd: command not found
ubuntu@ip-172-31-3-88:~/Node_application$ sudo firewall-cmd --add-service=http --permanent
sudo: firewall-cmd: command not found
ubuntu@ip-172-31-3-88:~/Node_application$ sudo firewall-cmd --reload
sudo: firewall-cmd: command not found
ubuntu@ip-172-31-3-88:~/Node_application$ npm install express

added 68 packages, and audited 69 packages in 3s

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  run `npm fund` for details

found 0 vulnerabilities
ubuntu@ip-172-31-3-88:~/Node_application$ node server.js &
[1] 7658
ubuntu@ip-172-31-3-88:~/Node_application$ Server running on http://localhost:3000
```

5) Configure Firewall and Security Group - Done

```
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15 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
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[1] 7658
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ubuntu@ip-172-31-3-88:~/Node_application$ ls
README.md  node_modules  package-lock.json  package.json  public  server.js
ubuntu@ip-172-31-3-88:~/Node_application$
```
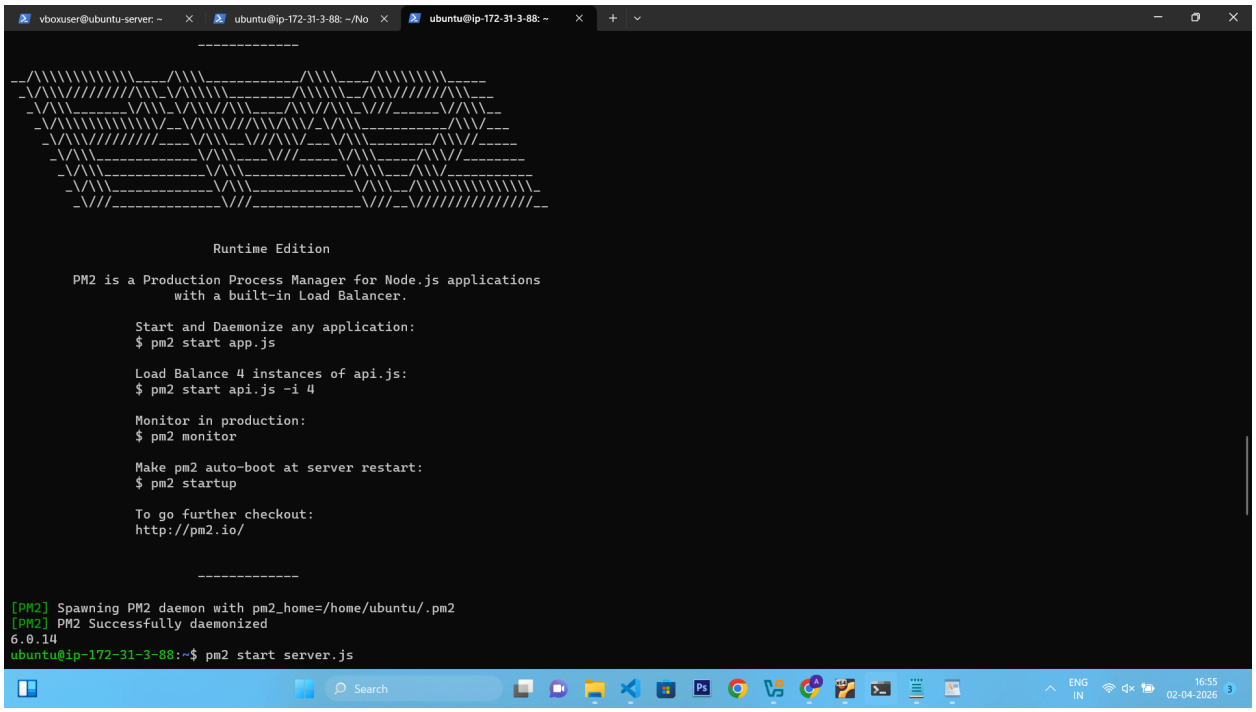
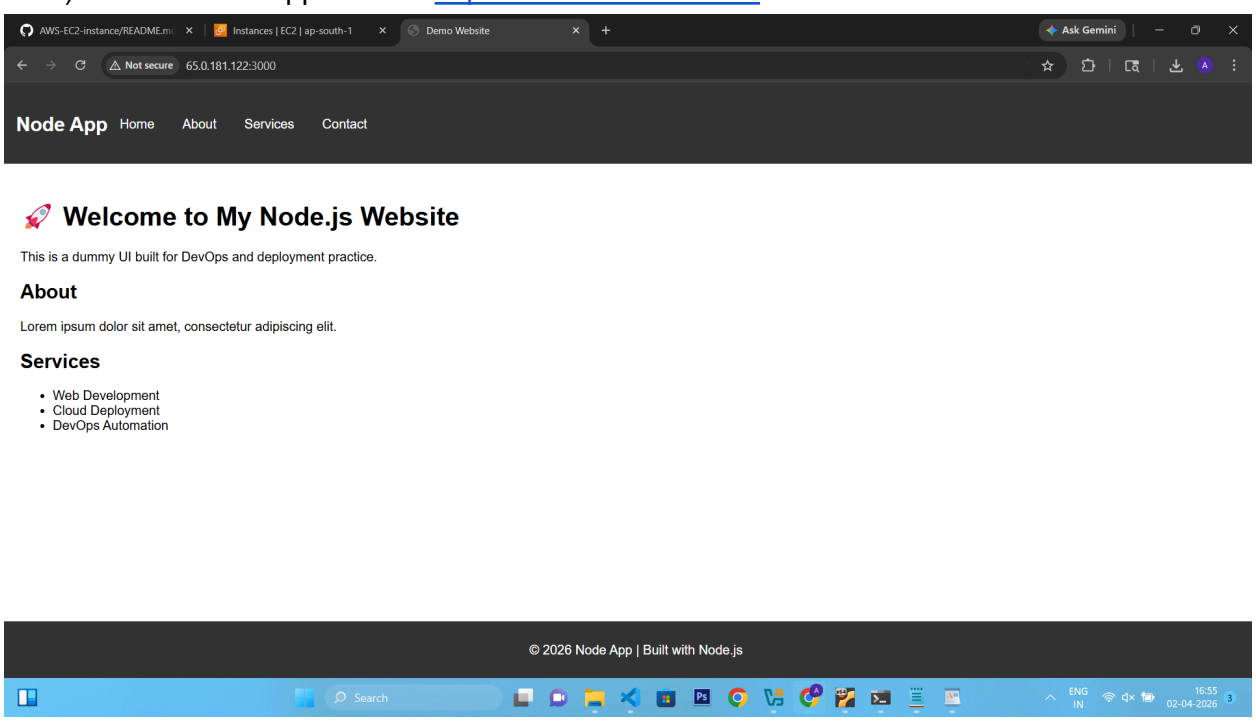

6) Set Up a Reverse Proxy (Optional): Done

7) Enable Auto-Start (Optional): Done



8) Access Your Application : http://65.0.181.122:3000/